

The Supply Edge: Where a Vacuum-Tied MOND Law Must Stop for Structure Growth and Galaxy Lensing to Agree

Working note

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Abstract

We work with the law $a_0 = \kappa c \sqrt{G \rho_\Lambda}$, $\kappa = \frac{1}{2}$ fitted, with the exponential kernel $\nu(y) = 1/(1 - e^{-\sqrt{y}})$, and a conserved cold fluid whose amount ($\Omega_c/\Omega_b = 5.364$) is an input. In particle-mesh simulations the law’s phantom over-builds small-scale structure: with the corrected pressure-filtered baryon source and a mass-conserving reservoir rule, the excess at $k \leq 1 \text{ h/Mpc}$ is $+15.3\%/+19.3\%$ (canonical/alternative footing) at 256^3 and grows to $+17.4\%$ at 512^3 . About 81% of the switch-ON mass behind it lies inside host turnaround spheres. Re-fitting KiDS-1000 isolated-lens lensing with a free two-halo amplitude shows the law need only extend to $x \simeq 0.3\text{--}0.5$ of the turnaround radius (best $x = 0.48/0.53$). We note that this is where a host’s real cold supply runs out: the phantom mass $M_{\text{ph}}(< r) = M_b/(e^s - 1)$, $s = r_t/r$, equals the supply $5.364 M_b/f_{\text{ret}}$ at $r_{\text{supply}} = r_t/\ln(1 + f_{\text{ret}}/5.364)$, which for census retention $f_{\text{ret}} = 0.07\text{--}0.10$ sits at $0.24\text{--}0.54 r_{\text{ta}}$ for KiDS lenses and $\simeq 0.24\text{--}0.33 r_{\text{ta}}$ for the groups and clusters that drive the excess. Inverting the relation gives a lensing-based cold-to-baryon ratio of $4.6\text{--}13.0$ (canonical; $6.0\text{--}16.5$ alternative), bracketing the CMB’s 5.364 on the canonical footing. We record this supply-edge hypothesis, its inputs and caveats (above all, the free two-halo term) and the pre-registered simulations that will test it.

1 Framework and status

The acceleration scale is $a_0 = \kappa c \sqrt{G \rho_\Lambda}$ with $\kappa = \frac{1}{2}$ fitted (two footings: 9.3603×10^{-11} and $1.1312 \times 10^{-10} \text{ m s}^{-2}$, never pooled). The baryon-sourced phantom is $\rho_{\text{ph}} = \nabla \cdot [(\nu - 1)g_b]/4\pi G$. A cold fluid of fixed cosmic amount is required; no particle species is added and the amount is not derived. In the working model the phantom is a target that real cold fluid must fill, and a host can fill it only as far as its supply allows.

2 The growth problem and where it lives

rule (particle mesh, $z = 0$, ratio to the Newtonian control)	σ_8 ratio	$\max P - 1 , k \leq 1$
phantom as extra gravity, max bookkeeping (256^3)	1.2049 / 1.2573	
reservoir rule + corrected gas filter (256^3)	1.0193 / 1.0243	0.153 / 0.193
the same, 512^3 (canonical)	1.0226	0.174

All are TENSION or worse against $|\sigma_8 - 1| \leq 5\%$ and $|P - 1| \leq 10\%$. In the 256^3 snapshots 0.805 of the switch-ON mass in filament-like cells lies inside resolved hosts’ turnaround spheres; a switch that stays ON out to r_{ta} removes at most 13–19% of the excess.

3 How far must the law extend for lensing?

An earlier fit required the law ON to r_{ta} because the two-halo amplitude was fixed at zero. With it free, KiDS-1000 isolated lenses prefer truncation at $x = 0.48$ (canonical) and 0.53 (alternative); $x = 0.3$ passes with any single bin dropped. In the nine bins the survey trusts, all x fit equally; the preference comes from outer bins where the free template carries much of the model. At $x = 0.4$, 0.83 of the excess source lies beyond the truncation in the simulations (an estimate, not a simulation result).

4 The supply edge

For a point mass with this kernel, $M_{\text{ph}}(< r) = M_b/(e^s - 1)$ with $s = r_t/r$, $r_t = \sqrt{GM_b/a_0}$. Setting it equal to the supply $5.364 M_b/f_{\text{ret}}$ gives

$$r_{\text{supply}} = \frac{r_t}{\ln(1 + f_{\text{ret}}/5.364)} \approx \frac{5.364}{f_{\text{ret}}} r_t,$$

equal to $5.850 r_t$ for $f_{\text{ret}} = 1$. With census retention, KiDS lenses ($\log M_b = 10.5\text{--}11.3$) have $x_{\text{edge}} = 0.24\text{--}0.54$ (canonical footing, small- f_{ret} form; about 10% lower on the alternative footing), and the simulations' resolved groups and clusters, with group/cluster retention, $x_{\text{edge}} = 0.24\text{--}0.33$. Both overlap the window that lensing allows and growth needs. If confirmed, the truncation radius is not a new constant but the point where the host's cold supply runs out.

5 A lensing estimate of the cold fluid's amount

Inverting, $R_c = x r_{\text{ta}} f_{\text{ret}}/r_M$ with KiDS's preferred x gives 4.6–13.0 on the canonical footing (containing 5.364) and 6.0–16.5 on the alternative footing. This is a consistency test of the supply-edge picture, not a derivation of the amount.

6 Pre-registered tests (running)

(i) The switch confined to $0.4 r_{\text{ta}}$ around resolved peaks at 512^3 ; (ii) the switch confined to each host's own supply edge with a declared retention-mass relation, at 512^3 ; (iii) the same with a_0 tracking DESI's evolving dark energy. Pass: $|\sigma_8 - 1| \leq 5\%$ and $|P - 1| \leq 10\%$ on both footings.

7 Caveats

The lensing window rests on a free two-halo amplitude with no external prior. Retention f_{ret} varies (7–18% in the census). The reservoir is gravity-level bookkeeping, the simulations do not model baryon depletion, the confinement's legality is conditional, and a separate consequence test of a Tolman reading of the factor 4 ($\kappa = 1/|1 + 3w_0|$) is not yet decisive with current DESI precision. $\kappa = \frac{1}{2}$ remains fitted; the cold fluid's amount and ρ_Λ are not derived.

Data and code

All scripts, frozen criteria and outputs are in the project repository (<https://github.com/carlzimmerman/zimmerman-formula>, folders `campaign_fresh_gravity/CFG361-CFG419` and `deepseek_push/openai_math_cross_analysis_2026-10`). AI assistance: Anthropic's Claude and DeepSeek wrote and ran code and drafted text; the author is responsible for the content.